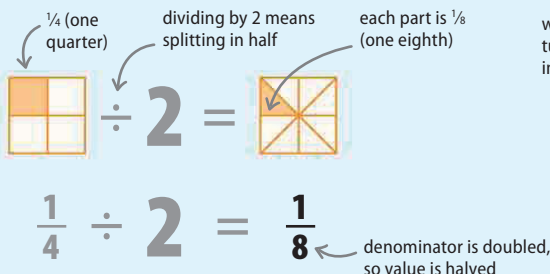
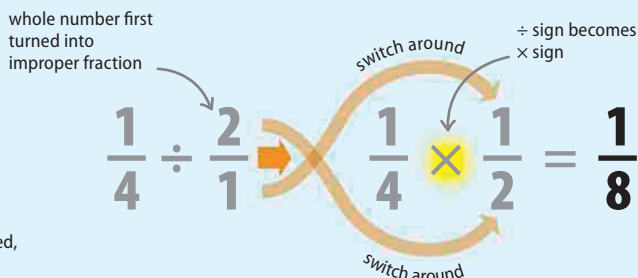


DIVIDING FRACTIONS

Fractions can be divided by whole numbers. Turn the whole number into a fraction and find the reciprocal of this fraction by turning it upside down, then multiply it by the first fraction.



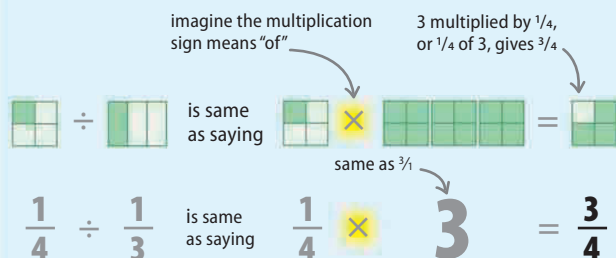
Picture dividing a fraction by a whole number as splitting it into that many parts. In this example, $\frac{1}{4}$ is split in half, resulting in twice as many equal parts.



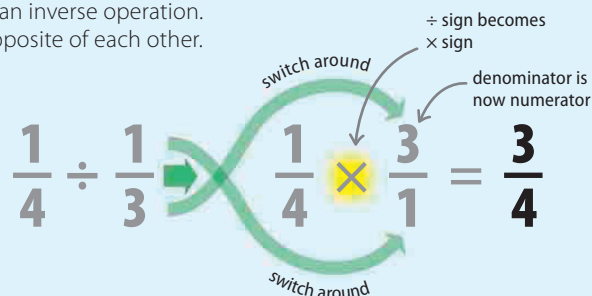
To divide a fraction by a whole number, convert the whole number into a fraction, turn that fraction upside down, and multiply both the numerators and the denominators.

Dividing two proper fractions

Proper fractions can be divided by other proper fractions by using an inverse operation. Multiplication and division are inverse operations—they are the opposite of each other.



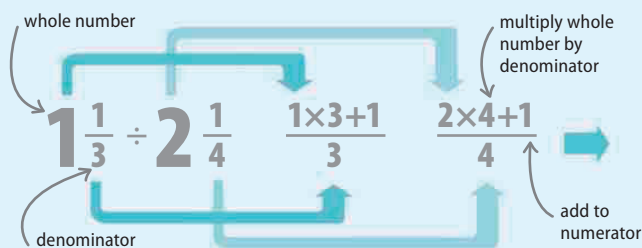
Dividing one fraction by another is the same as turning the second fraction upside down and then multiplying the two.



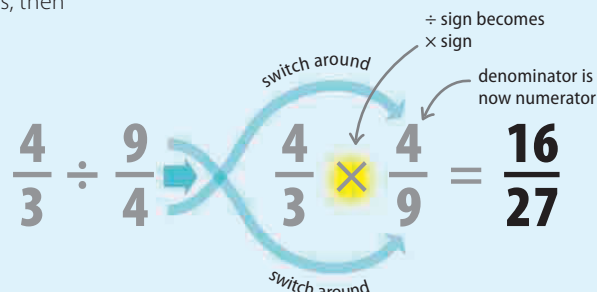
To divide two fractions use the inverse operation—turn the last fraction upside down, then multiply the numerators and the denominators.

Dividing mixed fractions

To divide mixed fractions, first convert them into improper fractions, then turn the second fraction upside down and multiply it by the first.



First, turn each of the mixed fractions into improper fractions by multiplying the whole number by the denominator and adding the numerator.



Divide the two fractions by turning the second fraction upside down, then multiplying the numerators and the denominators.